ASSESSMENT TOOL 1

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern. a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Event Communication Security Using Caesar Cipher

Aim :

To encrypt and decrypt a confidential message using the Caesar Cipher with a fixed shift value of **4**.

Theory :

The Caesar Cipher is one of the simplest classical substitution ciphers. In this technique, each letter of the plaintext is replaced by another letter that is a fixed number of positions ahead in the alphabet. The same shift value is used for both encryption and decryption. To recover the original message, the receiver shifts each encrypted letter backward by the same number of positions.

Algorithm :

Encryption

1. Take the plaintext message.
2. Choose the shift value (4).
3. Shift each alphabetic character 4 positions forward.
4. Keep spaces unchanged.
5. The resulting text is the ciphertext.

Decryption

1. Take the ciphertext.
2. Use the same shift value (4).
3. Shift each alphabetic character 4 positions backward.
4. Keep spaces unchanged.
5. The original plaintext is obtained.

a. <u>Encryption of the Message</u>

Plaintext:

MEET AT NOON

Shift Value: 4

Plaintext Letter	M	E	E	T	A	T	N	O	O	N
Shift (+4)	Q	I	I	X	E	X	R	S	S	R

Ciphertext:

QIIX EX RSSR

b. <u>Decryption of the Message</u>

The authorized member knows that the message was encrypted using a shift value of **4**.

Each letter of the ciphertext is shifted **4 positions backward** to recover the original message.

Ciphertext Letter	Q	I	I	X	E	X	R	S	S	R
Shift (−4)	M	E	E	T	A	T	N	O	O	N

Recovered Plaintext:

MEET AT NOON

<u>Explanation :</u>

The sender encrypts the message by shifting every alphabetic character 4 positions forward in the English alphabet. Since the authorized member knows the shift value (4), they decrypt the message by shifting every encrypted letter 4 positions backward, successfully recovering the original message.

<u>Result :</u>

The plaintext message **"MEET AT NOON"** was successfully encrypted into **"QIIX EX RSSR"** using a Caesar Cipher with a shift of 4. The original message was successfully retrieved by applying the reverse shift during decryption.

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of

the text suggests a simple substitution mechanism rather than complex encryption. a. Deduce the most likely original message. b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Intercepted Suspicious Message Using Atbash Cipher

Aim :

To analyze an intercepted coded message and determine its original plaintext using the Atbash Cipher.

Theory :

The **Atbash Cipher** is a classical monoalphabetic substitution cipher in which each letter of the alphabet is replaced with its corresponding letter from the opposite end of the alphabet. Unlike the Caesar Cipher, the Atbash Cipher does not use a shift value. It follows a fixed substitution pattern where **A is replaced by Z, B by Y, C by X**, and so on. Since the substitution is symmetrical, the same process is used for both encryption and decryption.

Algorithm :

Decryption

1. Take the intercepted ciphertext.
2. Replace each letter with its opposite letter in the alphabet.
3. Keep spaces unchanged.
4. Continue until all letters are converted.
5. The resulting text is the original plaintext.

a. Deducing the Original Message

Intercepted Ciphertext:

GSRH RH Z HVXIVG

Character-wise Decryption

Ciphertext Letter	**G**	**S**	**R**	**H**	**R**	**H**	**Z**	**H**	**V**	**X**	**I**	**V**	**G**
Plaintext Letter	T	H	I	S	I	S	A	S	E	C	R	E	T

Recovered Plaintext:

THIS IS A SECRET

b. Justification

The intercepted message exhibits characteristics of a **simple substitution cipher** rather than a complex encryption algorithm.

1. The repeated ciphertext pattern **"RH"** appears twice, which decrypts to **"IS"**, a common English word.
2. The single-letter ciphertext **"Z"** decrypts to **"A"**, the only valid single-letter English article.
3. The last six letters **"HVXIVG"** decrypt to **"SECRET"**, forming a meaningful English word.
4. The substitution follows a fixed reverse alphabet mapping (A↔Z, B↔Y, C↔X, …), which is the defining characteristic of the **Atbash Cipher**.
5. Since the same substitution rule is applied consistently throughout the message, the decrypted text forms the meaningful sentence:

THIS IS A SECRET

Hence, the message was most likely encrypted using the **Atbash Cipher**.

Result :

The intercepted ciphertext **"GSRH RH Z HVXIVG"** was successfully decrypted as **"THIS IS A SECRET"** using the **Atbash Cipher**. The repeating letter patterns, meaningful English words, and fixed reverse-alphabet substitution confirm that the message was encoded using this classical substitution cipher.